

PAPER_1201_UQFF_26D_Polynomial_Origami_Downward_Projection_Axiom

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PAPER_1201 -- UQFF 26D Polynomial Origami Downward Projection Axiom

Critical Rule (26D_DOWNWARD_PROJECTION.md:9) ``` ALL projections run DOWNWARD from 26D. 2D is a reference plane of observation -- NOT a foundation. WRONG: 2D -> 3D -> 9D -> 26D CORRECT: 26D -> 9D -> 3D -> 2D ```

Dimensional Hierarchy (26D_DOWNWARD_PROJECTION.md:22) ``` 26D -- ORIGIN UA pure energy, SCm injected -> DPM grinding pair (CW-SCm, CCW-UA') $F_{U^{26D}} = U_g + U_m + U_b + SCm/UA + BBDT$ * Prob_order $E^{26D} = UA + SCm * DPM_{ref} + BBDT$

9D -- VOID SYNTHESIS

$\det(M_{26 \rightarrow 9})$ compactification matrix (99 diagonal [d9,0,0 / ...])

$Void_{synth} = \det(M_{26 \rightarrow 9}) (U_g / U_m / U_b) / d3 + F_{inert} E^{26D} + QFP_{unique}$

$QFP_{unique} = \pi_{[n]} IG * Wolfram_{rules}$

3D -- OBSERVABLE MASS

$M = E^{26D} / c^{26} (1 - v_{current} / v_{init}) Prob_order$

c^{26} = speed-of-light equivalent in 26D manifold

2D -- REFERENCE PLANE ONLY

Observable slices (CMB, spectra) -- NOT building block

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**Core 26D Equations (26D\_DOWNWARD\_PROJECTION.md:54) ```  $F_{U^{26D}} = U_g + U_m + U_b + SCm/UA + BBDT * Prob\_order$   $M = E^{26D} / c^{26} * (1 - v_{current}/v_{init}) * Prob\_order$   $Void_{synth} = \det(M_{26 \rightarrow 9}) * (U_g/U_m/U_b)/d3 + F_{inert} * E^{26D} + QFP_{unique}$  Triple-root:  $x^3 - 3x^2 + 2x = 0 \rightarrow roots = [0, 1, 2]$  ```**

**26D Invariants in UQFF Axiom Set (\_uqff\_primitives.py:632) ``` \_s26\_3 = 1.4531e26 \_phi\_res = PHI\_RES \_n\_layers = 26 \_kappa = 0.0005 \_e0 = 1e-20 ```**

Usage in All 10 derive\_\* (\_uqff\_primitives.py:641) ``  
 derive\_alpha\_uqff: base = 1 / (PHI\_RES \* N\_LAYERS \* 2\pi) \* (1 + 0.001\*Ubi\_corr) Ubi\_corr = F\_TRZ \* RATIO / (S26\_3 \*\* (1/26))  
 derive\_G\_newton: lambda\_cross = (S26\_3\*\*(1/26)) \* (E0/rho)\*\*(1/3)  
 proj\_factor = (1 + beta\_i) / (N\_LAYERS \* (1 + F\_TRZ)) derive\_beta\_i:  
 cycle\_avg = 0.5 + 0.5\*cos(\pi \* F\_TRZ) kappa\_geom = (RATIO-1) \* (KAPPA \* N\_LAYERS) derive\_particle\_masses: omega\_base = THZ \* (rho\_scm/rho\_ua) \* (S26\_3\*\*(1/26)) \* beta\_i derive\_hbar: hbar\_base = (E0 \* S26\_3 \* PHI\_RES) / (c \* 26 \* 2\pi) \* ubi\_avg derive\_condensed...:  
 amp = S26\_3 \* PHI\_RES \* (1 + KAPPA\*1e4) derive\_habitable...: uses derive\_G + derive\_beta + derive\_rho (all 26D) ``

DPMVars26D (source172 + VerificationOrchestrator + 26D\_DOWNWARD\_PROJECTION.md:70) `` 19\*26 arrays:  
 f\_UA\_prime[19][26], f\_SCm[19][26] S26\_3, det(M\_26->9), fold signatures, downward projection invariants  
 evaluate\_26D\_polynomial(26D inputs) -> 9D/3D/2D outputs Used in: GeometricCheckpoint, QCalcGeom CP layers, all derive\_\*, 9-test Derivations Test ``

26D Polynomial in Quantum Chain Amplification  
 (dpm\_vacuum\_manifold.py:139) `` VDS = Sum( (SSQ\*\*n) / n\*\*26 , (n,1,oo) ) # = Li\_26([SSq]) E\_n = [E0 \* 10\*\*n for n in 1..26]  
 rho\_vac\_energy = sum(f\_SCm \* E for E in E\_n) / V ``

Cross-References - 26D\_DOWNWARD\_PROJECTION.md:11 (ALL projections downward from 26D) - \_uqff\_primitives.py:681  
 (lambda\_cross S26\_3^{1/26} in G) - dpm\_vacuum\_manifold.py:139 (VDS Li\_26) - QCalcGeom.py:33 (poly26\_derivative port) - LEDGER\_FIRST\_PRINCIPLES\_DERIVATIONS.md:150 (DPM layers = 26D polynomial origami) - WhitepaperGapAnalysis.md:100 (G3 gap closure)